IS101 — Introduction to Computing

SOC 101 Grade Book

College of Southern Nevada · Unit 3 — Spreadsheets · Variation 1 · 100 points · 26 tasks

How to use this document: Download the starter workbook from the Project Grader page. Open it in Microsoft Excel 365 and keep this sheet open beside it. Work through each task in order. When you're done, save the file and upload it to the grader for instant scoring.

Workbook Setup & Formatting Tasks 1–9

Task 1 3 points

Freeze the header row and the student-name column

WHY THIS MATTERS

Freezing keeps the headers and student names visible while scrolling through scores.

WHERE TO FIND IT

Click cell B2 (the first score cell) → View tab → Freeze Panes → Freeze Panes.

HOW YOU'LL KNOW IT WORKED

When you scroll down, the header row stays visible. When you scroll right, the name column stays visible.

Task 2 3 points

Add a comment on the curve column explaining the adjustment

WHY THIS MATTERS

Comments let you document decisions — like why a curve was applied — without changing the data.

WHERE TO FIND IT

Right-click any cell in the curve column → Insert Comment (or New Note) → type your explanation.

HOW YOU'LL KNOW IT WORKED

A red triangle appears in the cell corner; hovering shows your comment.

Task 3 2 points

Format the percentage columns with the Percent number format

WHY THIS MATTERS

Percent format displays decimal values as percentages (0.85 becomes 85%).

WHERE TO FIND IT

Select the percentage column data → Home tab → Number group → Percent Style (or Ctrl+Shift+5).

HOW YOU'LL KNOW IT WORKED

Values display as 85% instead of 0.85.

Task 4 3 points

Highlight failing scores (below 60) with conditional formatting

WHY THIS MATTERS

Conditional formatting visually flags students who need attention.

WHERE TO FIND IT

Select the score column → Home tab → Conditional Formatting → Highlight Cell Rules → Less Than → enter 60 → choose red fill → OK.

HOW YOU'LL KNOW IT WORKED

Cells with scores below 60 are highlighted in red.

Task 5 3 points

Name the grade-scale lookup range “GradeScale”

WHY THIS MATTERS

Named ranges make formulas easier to read. The grade scale table (with letter grades and cutoff scores) needs to be named so you can reference it in lookup formulas.

WHERE TO FIND IT

Select the grade scale range (including both the score cutoffs and letter grades) → click in the Name Box (left of the formula bar) → type GradeScale → press Enter.

HOW YOU'LL KNOW IT WORKED

The Name Box shows “GradeScale” when you select the range; Formulas {ARR} Name Manager lists it.

Task 6 3 points

Use the GradeScale named range inside a lookup formula

WHY THIS MATTERS

Referencing a named range in a formula makes it self-documenting — =VLOOKUP(score,GradeScale,2) is clearer than =VLOOKUP(score,$M$2:$N$6,2).

WHERE TO FIND IT

In a formula that looks up letter grades, type GradeScale instead of the cell range. Example: =VLOOKUP(F2,GradeScale,2,TRUE).

HOW YOU'LL KNOW IT WORKED

The formula bar shows a formula containing the word “GradeScale” instead of cell references.

Task 7 2 points

Wrap text in the instructor-feedback column

WHY THIS MATTERS

Wrap Text shows long feedback on multiple lines within the cell instead of cutting it off.

WHERE TO FIND IT

Select the feedback column → Home tab → Alignment group → Wrap Text.

HOW YOU'LL KNOW IT WORKED

Long feedback text wraps to multiple lines within each cell.

Task 8 3 points

Convert the grade data to an Excel Table

WHY THIS MATTERS

Excel Tables add automatic filtering, banded rows, and structured references.

WHERE TO FIND IT

Click anywhere in the grade data → press Ctrl+T (or Insert tab → Table) → verify the range → OK.

HOW YOU'LL KNOW IT WORKED

Filter arrows appear on each column header; the Table Design tab shows when you click in the table.

Task 9 3 points

Sort by final grade (descending) then last name (ascending)

WHY THIS MATTERS

Multi-level sorting puts the highest grades first and alphabetizes students with the same grade.

WHERE TO FIND IT

Data tab → Sort → Sort by: Final Grade → Order: Largest to Smallest → Add Level → Then by: Last Name → Order: A to Z → OK.

HOW YOU'LL KNOW IT WORKED

Students with the highest grades appear first; students with equal grades are sorted alphabetically.

Formulas & Functions Tasks 10–18

Task 10 6 points

Use SUM and AVERAGE to total and average each student's scores

WHY THIS MATTERS

SUM adds up all scores; AVERAGE calculates the mean. Both are essential for a grade book.

WHERE TO FIND IT

Click in the total column for the first student → type =SUM(and select the score cells → Enter. In the average column, type =AVERAGE(and select the same range → Enter. Copy both formulas down for all students.

HOW YOU'LL KNOW IT WORKED

Each student has a total and an average calculated from their individual scores.

Task 11 4 points

Use COUNT or COUNTA to tally submitted assignments per student

WHY THIS MATTERS

COUNT counts numeric cells; COUNTA counts non-empty cells. Either works to count how many assignments a student submitted.

WHERE TO FIND IT

Click in the "Submitted" column for the first student → type =COUNTA(and select the score cells → Enter. Copy down for all students.

HOW YOU'LL KNOW IT WORKED

Each student shows a count of how many assignments they submitted.

Task 12 4 points

Use an absolute reference ($) in the percentage formula

WHY THIS MATTERS

Absolute references (with $) lock a cell reference so it doesn't shift when you copy a formula. Use it to reference the total-points cell.

WHERE TO FIND IT

In the percentage formula, reference the total-points cell with $ signs: e.g., =F2/$F$1. The $ before F and 1 keeps it locked when you copy down.

HOW YOU'LL KNOW IT WORKED

The formula bar shows $ signs in the reference; copying the formula down keeps the same denominator cell.

Task 13 5 points

Use IF to mark each student Pass or Fail

WHY THIS MATTERS

IF tests a condition and returns one value if true, another if false.

WHERE TO FIND IT

Click in the Pass/Fail column → type =IF(average>=60,"Pass","Fail") using the student's average cell → Enter. Copy down.

HOW YOU'LL KNOW IT WORKED

Each student shows "Pass" or "Fail" based on their average.

Task 14 6 points

Use a nested IF to assign letter grades A–F

WHY THIS MATTERS

A nested IF tests multiple conditions in sequence: =IF(score>=90,"A",IF(score>=80,"B",IF(score>=70,"C",IF(score>=60,"D","F")))).

WHERE TO FIND IT

Click in the letter grade column → type the nested IF formula using each student's average → Enter. Copy down.

HOW YOU'LL KNOW IT WORKED

Each student shows a letter grade (A, B, C, D, or F) based on their average.

Task 15 4 points

Use COUNTIF to count how many students earned an A

WHY THIS MATTERS

COUNTIF counts cells that match a condition — useful for class statistics.

WHERE TO FIND IT

Click in a summary cell → type =COUNTIF(grade_column,"A") → Enter.

HOW YOU'LL KNOW IT WORKED

The cell shows the number of students who earned an A.

Task 16 6 points

Use VLOOKUP against the grade scale to return each letter grade

WHY THIS MATTERS

VLOOKUP searches the GradeScale table for each student's score and returns the corresponding letter grade.

WHERE TO FIND IT

Click in the VLOOKUP grade column → type =VLOOKUP(average,GradeScale,2,TRUE) → Enter. Copy down.

HOW YOU'LL KNOW IT WORKED

Each student shows a letter grade looked up from the grade scale table.

Task 17 5 points

Wrap the VLOOKUP in IFERROR to handle errors gracefully

WHY THIS MATTERS

IFERROR catches errors (like #N/A) and shows a friendly message instead. Both IFERROR and VLOOKUP must be in the same cell.

WHERE TO FIND IT

Modify the VLOOKUP formula to: =IFERROR(VLOOKUP(average,GradeScale,2,TRUE),"Check") → Enter. Copy down.

HOW YOU'LL KNOW IT WORKED

Students with valid scores show letter grades; any errors show "Check" instead of #N/A.

Task 18 3 points

Use TODAY() to stamp the report-generated date

WHY THIS MATTERS

TODAY() inserts the current date and updates automatically each time the file opens.

WHERE TO FIND IT

Click in a cell near the top of the sheet (e.g., next to "Report Date:") → type =TODAY() → Enter.

HOW YOU'LL KNOW IT WORKED

The cell shows today's date and updates whenever you open the file.

Charts Tasks 19–21

Task 19 4 points

Create a Column chart of class average by assignment

WHY THIS MATTERS

Charts visualize how the class performed on each assignment.

WHERE TO FIND IT

Select the assignment names and class averages → Insert tab → Charts group → Column → Clustered Column.

HOW YOU'LL KNOW IT WORKED

A column chart appears showing average scores for each assignment.

Task 20 4 points

Add data labels and a chart title

WHY THIS MATTERS

Data labels show exact values on each bar; a chart title identifies what the chart represents.

WHERE TO FIND IT

Click the chart → Chart Design tab → Add Chart Element → Data Labels → Outside End. Then Add Chart Element → Chart Title → Above Chart → type your title.

HOW YOU'LL KNOW IT WORKED

Each bar shows its value; the chart has a descriptive title at the top.

Task 21 3 points

Add alt text to the chart

WHY THIS MATTERS

Alt text describes the chart for screen readers and accessibility compliance.

WHERE TO FIND IT

Right-click the chart → Edit Alt Text → type a description (e.g., “Column chart showing class average scores by assignment”).

HOW YOU'LL KNOW IT WORKED

Right-clicking the chart and selecting Edit Alt Text shows your description.

Finishing Touches Tasks 22–26

Task 22 3 points

Bold and shade the header row

WHY THIS MATTERS

Bold text and a colored background make the header row stand out from the data.

WHERE TO FIND IT

Select the header row → Home tab → Bold (Ctrl+B) → then click Fill Color and choose a color.

HOW YOU'LL KNOW IT WORKED

The header row has bold text on a colored background.

Task 23 4 points

Add a header with the course name and a footer with page number

WHY THIS MATTERS

Headers and footers identify printed pages.

WHERE TO FIND IT

Page Layout tab → Page Setup dialog launcher → Header/Footer tab → Custom Header: type “SOC 101 Grade Book” → Custom Footer: insert page number.

HOW YOU'LL KNOW IT WORKED
Print Preview shows the course name at the top and page number at the bottom.

Task 24 3 points

Set Title and Author workbook properties

WHY THIS MATTERS
Metadata makes the workbook identifiable and searchable.

WHERE TO FIND IT
File tab → Info → Properties panel. Set Title to “SOC 101 Grade Book” and Author to your name.

HOW YOU'LL KNOW IT WORKED
File {ARR} Info shows Title and Author populated.

Task 25 3 points

Filter to students currently failing

WHY THIS MATTERS
Filtering shows only students who need intervention.

WHERE TO FIND IT
Click the dropdown arrow on the Grade column header → uncheck Select All → check only “F” → OK.

HOW YOU'LL KNOW IT WORKED
Only students with an F grade are visible.

Task 26 8 points

Center the score columns

WHY THIS MATTERS
Centering aligns numeric scores in the middle of each cell for a clean, professional look.

WHERE TO FIND IT
Select all score columns → Home tab → Alignment group → Center.

HOW YOU'LL KNOW IT WORKED
All score values are centered in their cells.

Total: 100 points. Every task is graded automatically when you upload your file. © Dr. Mark Taormino · All Rights Reserved · College of Southern Nevada